Fiscal Year:	FY 2020	**Task Last Updated:**	FY 06/29/2020
PI Name:	Riccio, Gary Ph.D.		
Project Title:	Translational Research for Autonomous Care Coordination (TRACC)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02108	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Synergy
Start Date:	03/01/2019	**End Date:**	02/29/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Menon, Jayant M.D. (Nahlia Inc.) Shelhamer, Mark Sc.D. (Johns Hopkins Medical School)		
Grant/Contract No.:	NNX16AO69A-SYN0007		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Synergy Project This project aims to provide human factors augmentation for the Translational Research Institute for Space Health (TRISH) funded project Autonomous Medical Response Agent (Principal Investigator: Jayant Menon) by assessing consequences that medical conditions have for resilience of exploration capabilities as well as the health hazards imposed by those capabilities.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Interdependencies Among Gaps Across Independent Lines of Research on Human-Systems Risks Med-10 (computed medical decision support) and Med-03 (personalized medicine) were viewed as inextricably connected to the gap for a capability to predict estimated medical risk posture during exploration missions based on current crew health and resources (Med-09). This may become the most important gap in the entire Human Research Program (HRP) portfolio to the extent that all risks are not burned down to the level of optimization by the time irrevocable decisions about exploration systems must be made and by the time a crew begins its journey beyond low-Earth orbit. The focus on preparation for adaptive resilience as the mission unfolds emphasizes practical interdependencies with a number of other gaps across risks--as well as the respective HRP Elements and associated stakeholders--including Med04, 05, 07, 08; BMed2, 3, 6, 7; HARI-01, 02, 03, 04; Train-01, 02, 03; Task-01, 02, 03; and Team1, 5, 6 (Ed. note: see Human Research Roadmap for gap names: https://). Multi-Stakeholder Coordination Across Technical Authorities in NASA for Demand-Side Innovation. The research in TRACC has revealed that the capabilities of Autonomous Medical Response Agent (AMRA) have direct implications for stakeholders who must address the downstream consequences of in-flight medical conditions with respect to (a) departures and responsible assets articulated in the requirements of NASA Standard (STD)- 3001 v2 for autonomous system; (b) prioritization of exploration tasks given their importance to mission success, disability and productivity of crew, and difficulty of adaptation and relearning by the crew; (c) dynamic re-planning and re-scheduling by crew and asynchronous coordination with the ground support team of teams; tradeoffs in physiological and medical status monitoring; (d) multi-scale relationships among biological, psychological, and behavioral measures; (e) meta-scientific model-based management of heterogeneous archives for clinical and scientific data; and (f) resilience engineering for human-systems interaction design of complex systems. Development and Promulgation of NASA's Metascience for both Space and Terrestrial Applications. The most important take-away points from the NASA's human research correspond to six meta-scientific values around which scientific and technical innovation as well as associated organizational change initiatives can be organized, most notably for digitization and consumerization of healthcare in a smart medical home: 1. Meta-scientific frameworks and forums for multi-stakeholder disintermediation; 2) Managing scope and detail of research through design reference missions; 3) Diversity of experimental analogs for holistic translational research; 4) Assessment of consequences of medical status for life and work; 5) Multi-scale computational modeling, simulation, and analysis; 6) Scientific focus on use, utility, and value at the point of care. Surge Capacity for Pandemics. An extensive review and holistic approach was described for knowledge transfer that builds on NASA's metascience to develop the capacity to coordinate the operation of a system of systems by a distributed team of teams--amid variability in connectivity, bandwidth, time delay ,and sign-to-noise ratio--to maintain health and well-being of more or less isolated individuals to support both life and work in potentially austere environments who may suffer from a wide variety of medical conditions. Austere environments for both life and work almost always involve assistance from people who may not be experts in the necessary subject matter and who may not have access to technology and products specifically designed for the medical condition at hand. A decentralized system of systems that utilizes a smart medical home can accommodate such situations and thus would be invaluable in a pandemic in which the capacity of standing healthcare institutions is exceeded potentially by an order of magnitude.
Task Progress:	The intent of Translational Research for Autonomous Care Coordination (TRACC) was to enhance the use, utility, and value of an Autonomous Medical Response Agent (AMRA) for a variety of exploration stakeholders. The research focused on opportunities to address needs implicit in HRP's red risk of Adverse Health Outcomes & Decrements in Performance due to Inflight Medical Conditions. At the outset of the TRACC project, investigators utilized an Agile Scrum methodology to identify the most important stories in the development of AMRA, specifically as they relate to collectively unique and individually essential attributes of the innovation. Stories were bundled across levels of system decomposition in the system of systems for Exploration Medical Capabilities. Bundles then were prioritized based on the potential magnitude of impact of TRACC on the ongoing development of AMRA. Sources of evidence in NASA were identified for each of the stories, due diligence on which then became the basis for tasking for each story. Deliverables outlined in the proposal were addressed though the work on various tasks within each story. Findings and conclusions from these tasks were summarized in a monograph (over 40,000 words) with approximately 300 reference citations, entitled Situated Precision Healthcare in the Smart Medical Home. A key finding is that a response to the HRP gap for computed medical decision support during exploration missions (Med-10) should be intertwined with a broader interpretation and response to the gap for personalized medicine (Med-03) that can accommodate the need for near-real-time assessment beyond omics. The approach to personalized medicine focused on delivering collaborative decision support that incorporates rich outcomes-based constraints revealed by multi-scale qualitative and quantitative health observations that evolve for individuals and crews in real time during a mission. Another key finding is that Med-10 and Med-03 are inextricably connected to the gap for a capability to predict estimated medical risk posture during exploration missions based on current crew health and resources (Med-09). This may become the most important gap in the entire HRP portfolio to the extent that all risks are not burned down to the level of optimization by the time irrevocable decisions about exploration systems must be made and by the time a crew begins its journey beyond low-Earth orbit. The focus on preparation for adaptive resilience as the mission unfolds emphasizes practical interdependencies with a number of other gaps across risks and responsible HRP Elements including Med-04, 05, 07, 08; BMed-02, 03, 06, 07; HARI-01, 02, 03, 04; Train-01, 02, 03; Task-01, 02, 03; and Team-01, 05, 06. Our research in TRACC has revealed that the capabilities of AMRA have direct implications for stakeholders who must address the downstream consequences of in-flight medical conditions with respect to (a) departures and responsible assets articulated in the requirements of NASA STD- 3001 v2 for autonomous system; (b) prioritization of exploration tasks given their importance to mission success, disability and productivity of crew, and difficulty of adaptation and relearning by the crew; (c) dynamic re-planning and re-scheduling by crew and asynchronous coordination with the ground support team of teams; tradeoffs in physiological and medical status monitoring; (d) multi-scale relationships among biological, psychological, and behavioral measures; (e) meta-scientific model-based management of heterogeneous archives for clinical and scientific data; and (f) resilience engineering for human-systems interaction design of complex systems.
Bibliography Type:	Description: (Last Updated: 07/29/2020)

Abstracts for Journals and Proceedings	Riccio G, Shelhamer M, Menon J. "Synergy between developers and stakeholders for autonomous medical decision making." 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020
Articles in Peer-reviewed Journals	Riccio G, Au R, van Emmerik R, Eslami M. "Situated precision health in the smart medical home: Bringing NASA's research strategy down to Earth." Adv Geriatr Med Res. 2020;2(3):e200017. Published: 29 June 2020. https://doi.org/10.20900/agmr20200017 , Jun-2020